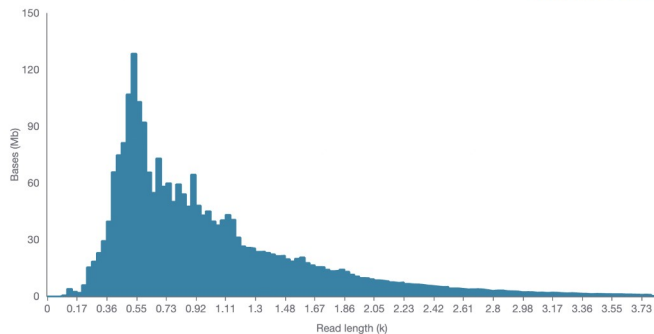
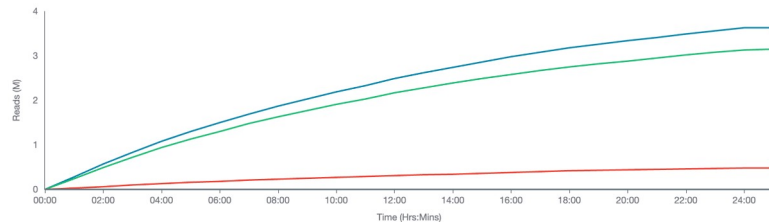
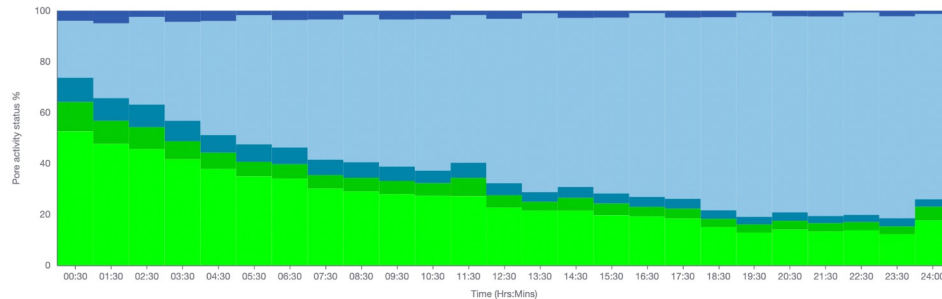


Sequencing run QC for ONT



*N50 calculated from basecalled read length histogram.



What is covered in this section?

Run output files

Sequencing run report

Next steps

Output files generated from ONT run

Run Folder:

Name	
 barcode_alignment_FAY32108_7a4b3362_379472e5.tsv	
>  fastq_fail	←
>  fastq_pass	
 final_summary_FAY32108_7a4b3362_379472e5.txt	
>  other_reports	
>  pod5	←
 pore_activity_FAY32108_7a4b3362_379472e5.csv	
 report_FAY32108_20240813_1256_7a4b3362.html	←
 report_FAY32108_20240813_1256_7a4b3362.json	
 report_FAY32108_20240813_1256_7a4b3362.md	
 sample_sheet_FAY32108_20240813_1256_7a4b3362.csv	
 sequencing_summary_FAY32108_7a4b3362_379472e5.txt	
 throughput_FAY32108_7a4b3362_379472e5.csv	

Basecalled fastq files

(fail = Q-score is below threshold set during sequencing run (9 by default))

Raw sequencing files

Summary report for doing QC

Output fastq file organization

- Reads are automatically demultiplexed during basecalling
- Fastq files are generated in real-time (every 10 minutes in this example), so many fastq files are generated for each barcode (sample)

 barcode_al...379472e5.tsv 	 barcode01  >	 FAY32108_p...5_0.fastq.gz  Aug 13, 2024 at 1:06 PM
 fastq_fail  >	 barcode02  >	 FAY32108_p...5_1.fastq.gz  Aug 13, 2024 at 1:16 PM
 fastq_pass  >	 barcode03  >	 FAY32108_p...5_2.fastq.gz  Aug 13, 2024 at 1:26 PM
 final_summa...79472e5.txt 	 barcode04  >	 FAY32108_p...5_3.fastq.gz  Aug 13, 2024 at 1:36 PM
 other_reports  >	 barcode05  >	 FAY32108_p...5_4.fastq.gz  Aug 13, 2024 at 1:46 PM
 pod5  >	 barcode06  >	 FAY32108_p...5_5.fastq.gz  Aug 13, 2024 at 1:56 PM
 pore_activity...9472e5.csv 	 barcode07  >	 FAY32108_p...5_6.fastq.gz  Aug 13, 2024 at 2:06 PM
 report_FAY3...4b3362.html 	 barcode08  >	 FAY32108_p...5_7.fastq.gz  Aug 13, 2024 at 2:16 PM
 report_FAY3...4b3362.json 	 barcode09  >	 FAY32108_p...5_8.fastq.gz  Aug 13, 2024 at 2:26 PM
 report_FAY3...a4b3362.md 	 barcode10  >	 FAY32108_p...5_9.fastq.gz  Aug 13, 2024 at 2:36 PM
 sample_she...a4b3362.csv 	 barcode11  >	 FAY32108_p...10.fastq.gz  Aug 13, 2024 at 2:46 PM
 sequencing...379472e5.txt 	 barcode12  >	 FAY32108_p...5_11.fastq.gz  Aug 13, 2024 at 2:56 PM
 throughput_...79472e5.csv 	 barcode13  >	 FAY32108_p...12.fastq.gz  Aug 13, 2024 at 3:06 PM
	 barcode14  >	 FAY32108_p...13.fastq.gz 
	 barcode15  >	 FAY32108_p...14.fastq.gz 

Sequencing run QC using the “report_***.html” file

Run Folder:

Name	
	barcode_alignment_FAY32108_7a4b3362_379472e5.tsv
> 	fastq_fail
> 	fastq_pass
	final_summary_FAY32108_7a4b3362_379472e5.txt
> 	other_reports
> 	pod5
	pore_activity_FAY32108_7a4b3362_379472e5.csv
	report_FAY32108_20240813_1256_7a4b3362.html
	report_FAY32108_20240813_1256_7a4b3362.json
	report_FAY32108_20240813_1256_7a4b3362.md
	sample_sheet_FAY32108_20240813_1256_7a4b3362.csv
	sequencing_summary_FAY32108_7a4b3362_379472e5.txt
	throughput_FAY32108_7a4b3362_379472e5.csv

Summary report for doing QC

Amount of Data Generated

- Check that the amount of data generated matches your expectation (this is a function of library loading concentration, run time, number of pores available for sequencing)
- For a fresh flow cell with a run that's 48+ hours, ONT claims ~20 Gbases of data will be generated, but in practice, this number is ~1-6 Gbases
- If the flow-cell is being re-used and has <800 pores left (like example below), data generated will be lower

Run summary

DATA OUTPUT

Estimated bases

3.09 Gb

Reads generated

3.63 M

Estimated N50

860 b

BASECALLING

Reads called

100%

Reads called (min Q score: 9)

3.15 M

480.34 k

Pass

Fail

Bases called (min Q score: 9)

2.44 Gb

347.16 Mb

Pass

Fail

RUN DURATION

Run time

24 hrs 0 mins / 24 hrs 0 mins

Elapsed time



Run limit



Run status

FINISHED · Target runtime has been reached

Read length distribution

- Check that the distribution of read lengths matches expectation (depends on quality of sample going in, whether fragmentation was performed, etc.)
- Distribution should roughly match the library size seen in QSep/Tapestation/diagnostic PCR, although it's common for read lengths to skew shorter

^ READ LENGTHS · OUTLIERS REMOVED

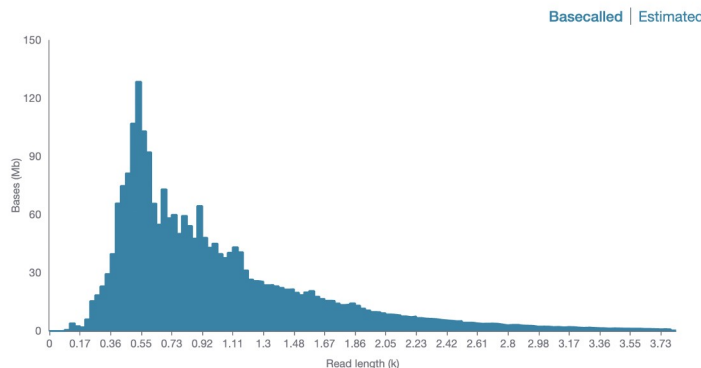
The read length graph shows the total number of bases vs the read length. The longest 1% of strands are classified as outliers, and excluded to allow focus on the main body of data.

N50*

860 b

% Basecalled

100 %



*N50 calculated from basecalled read length histogram.

Cumulative output as a function of time

- Cumulative output should start out linear and then plateau off as pores get exhausted
- >80% of bases should pass the Q-score threshold (usually set to 9)

Reads

Legend

— Total

Total number of reads, including passed, failed and skipped.

— Passed

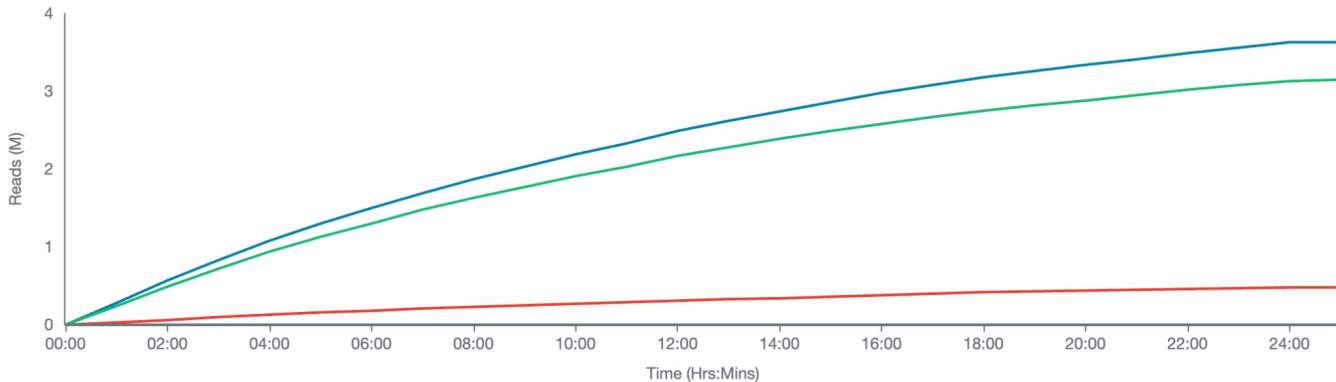
Reads equal to or above the quality score threshold.

— Failed

Reads below the quality score threshold.

— Skipped

Reads that will not be basecalled. Post run basecalling is possible.



Cumulative output as a function of time

- Cumulative output should start out linear and then plateau off as pores get exhausted
- >80% of bases should pass the Q-score threshold (usually set to 9)

Reads

Legend

— Total

Total number of reads, including passed, failed and skipped.

— Passed

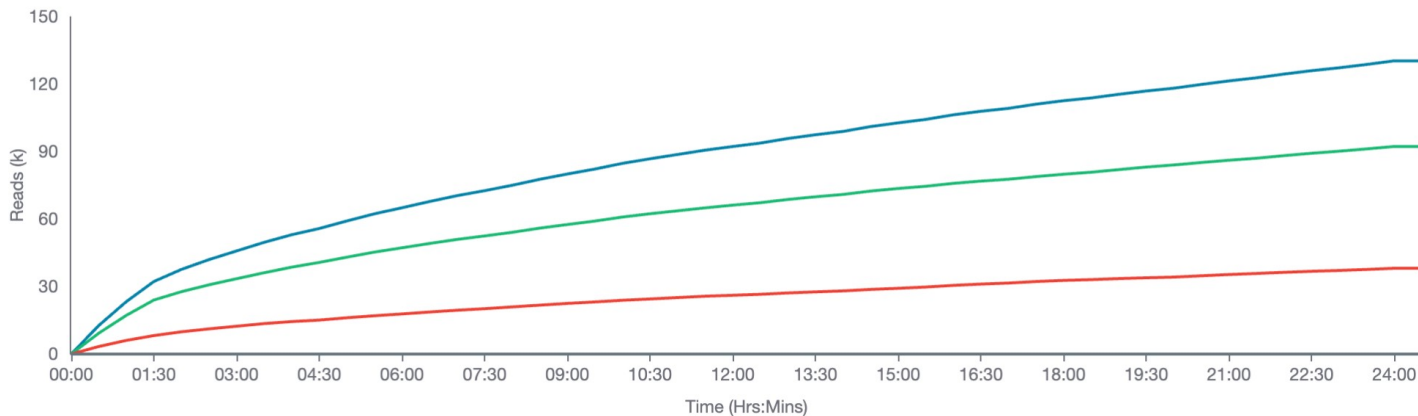
Reads equal to or above the quality score threshold.

— Failed

Reads below the quality score threshold.

— Skipped

Reads that will not be basecalled. Post run basecalling is possible.



Cumulative output as a function of time

- Cumulative output should start out linear and then plateau off as pores get exhausted
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Reads

Legend

— Total

Total number of reads, including passed, failed and skipped.

— Passed

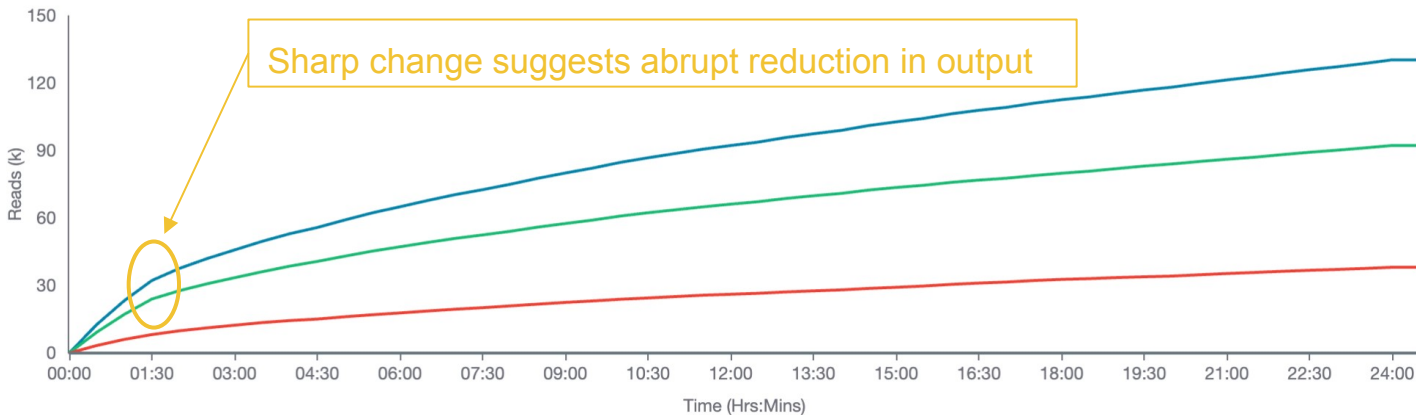
Reads equal to or above the quality score threshold.

— Failed

Reads below the quality score threshold.

— Skipped

Reads that will not be basecalled. Post run basecalling is possible.



Distribution of reads/bases across barcodes

- Check that this matches expectation - we generally aim for even distribution across barcodes*

Detected barcodes

The total number of bases and reads for each barcode detected are displayed in table below. Reads/bases must have a quality score above 9 to pass.

Unclassified data

416.02 k (13.2%)

350.51 Mb (14.4%)

Passed Reads

Passed Bases

Search barcodes

[Export CSV](#)

Barcode	Total bases (Mb)	Passed bases (%)	Total reads (k)	Passed reads (%)
barcode13	231.11	94.4	238.15	94.7
barcode14	168.85	96.1	166.23	96
barcode15	202.46	97	176.72	96.2
barcode16	1.04	77	3.82	75.5
barcode17	278.62	94.1	399.89	93.9
barcode18	204.88	94.7	296.47	94
barcode19	103.68	94.7	150.67	93.2
barcode20	2.07	84.4	5.3	74.3
barcode21	196.7	94.7	226.71	94.3
barcode22	144.67	95.3	170.5	94.2
barcode23	4.66	93	6.56	85.1
barcode24	0.62	70.4	2.37	64.6

*When setting up a run, there is the option (beta) to do barcode balancing by not sequencing overrepresented barcodes, but this hasn't been tested by RR team

Distribution of reads/bases across barcodes

- Check that this matches expectation - we generally aim for even distribution across barcodes*

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[Export CSV](#)

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barcode17	278.62	94.1	399.89	93.9
barcode18	204.88	94.7	296.47	94
barcode19	103.68	94.7	150.67	93.2
barcode20	2.07	84.4	5.3	74.3
barcode21	196.7	94.7	226.71	94.3
barcode22	144.67	95.3	170.5	94.2
barcode23	4.66	93	6.56	85.1
barcode24	0.62	70.4	2.37	64.6

Use this to evaluate
your sample
normalization /
pooling

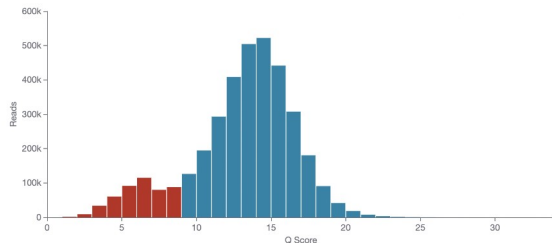
*When setting up a run, there is the option (beta) to do barcode balancing by not sequencing overrepresented barcodes, but this hasn't been tested by RR team

Q-score

- A mean score of 15 is typical for HAC basecalled runs
 - Q-score depends on the basecaller used
- Ideally, Q-scores will remain steady over time

Q Score histogram

■ Passed simplex reads
■ Failed simplex reads



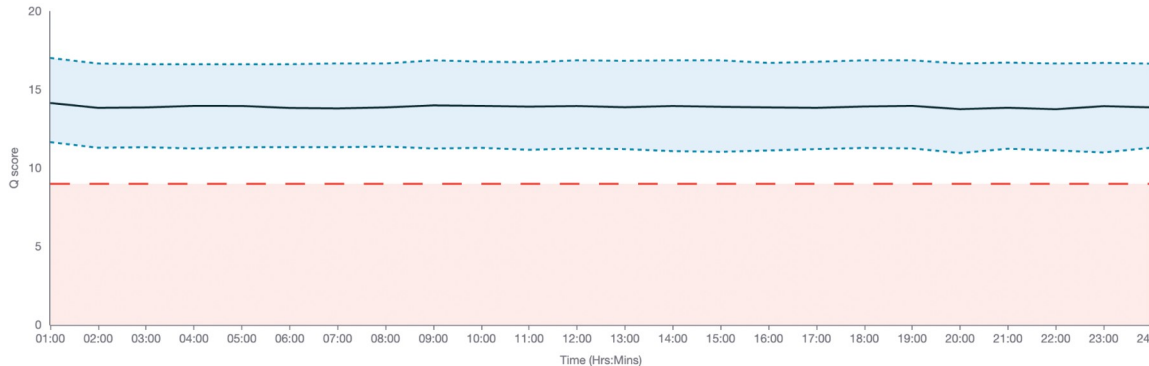
Q Score over time

Legend

— Mode
The most frequent quality score of reads in the run.

■ Spread
The spread of quality scores, found by calculating full width half maximum.

--- Min. quality score
Minimum quality score to be accepted as a passed read.



Q-score

- A mean score of 15 is typical for HAC basecalled runs
 - Q-score depends on the basecaller used
- Ideally, Q-scores will remain steady over time

^ QUALITY SCORE

The quality score is calculated as basecalling is performed on your device. Reads that fall below the minimum value of 9 will be classified as failed reads. You can alter the accepted minimum quality score in MinKNOW.

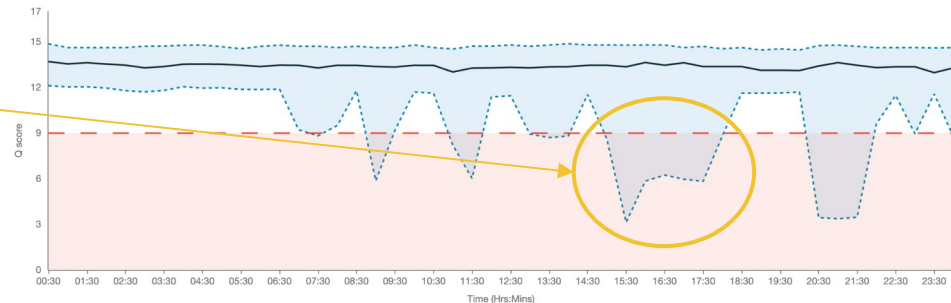
Legend

— Mode
The most frequent quality score of reads in the run.

□ Spread
The spread of quality scores, found by calculating full width half maximum.

--- Min. quality score
Minimum quality score to be accepted as a passed read.

Greater variation (reduction) in Q-score at later times



Pore activity over time

- Sequencing/available pores gradually decrease over time, with slight increases when pore scans are performed

Run health

PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

☒ Show grouped

Legend

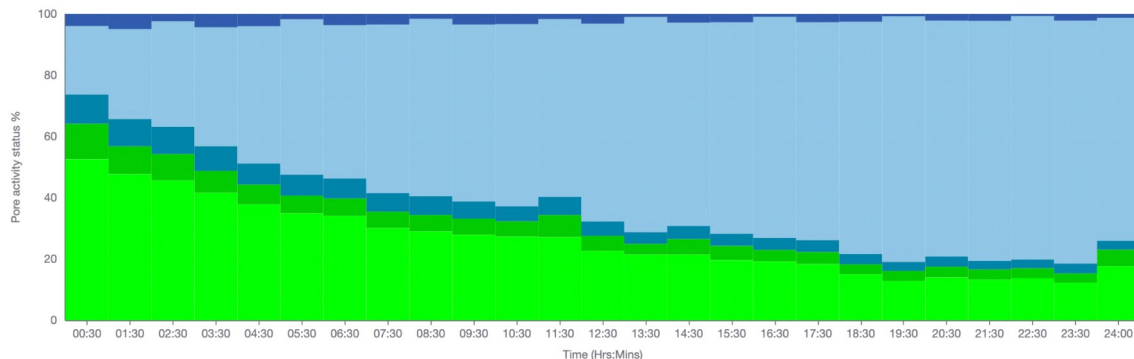
■ Sequencing
Pore currently
sequencing

■ Pore available
Pore currently
available for
sequencing

■ Unavailable
Pore currently
unavailable for
sequencing

■ Inactive
Pore no longer suitable
for further sequencing

■ Unclassified
Pore status unknown



Pore activity over time

- See [here](#) for how to diagnose issues based on pore activity plots

PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

Legend

Sequencing
Pore currently sequencing

Adapter
Pore currently sequencing adapter

Pore available
Pore available for sequencing

Unavailable
Pore unavailable for sequencing

Active feedback
Channel ejecting analyte

No pore
No pore detected in channel

Out of range-high
Current is positive but unavailable for sequencing

Out of range-low
Current is negative but unavailable for sequencing

Multiple
Multiple pores detected. Unavailable for sequencing.

Saturated
The channel has switched off as current levels exceed hardware limitations

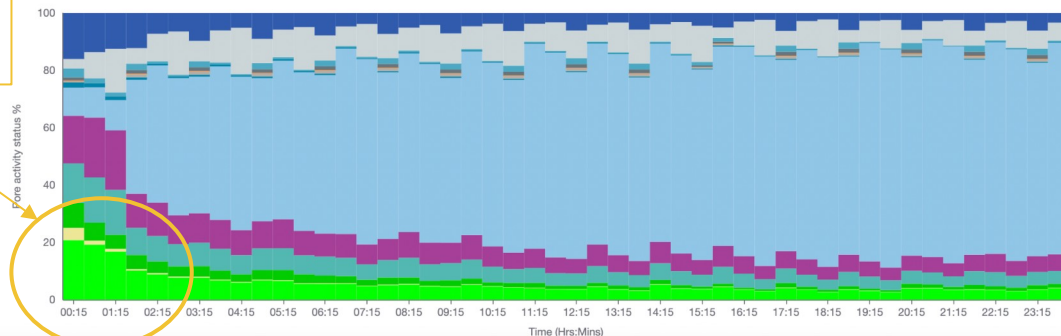
Zero
Pore currently unavailable for sequencing

Channel disabled
Channel is disabled and awaiting another pore scan

Unclassified
Pore status unknown

Lots of unusual pore statuses

Low starting % of active pores



Pore activity over time can also help indicate if the right amount of library was loaded

- **Was the right amount of library loaded?**
- We aim for a high % of pores sequencing (light green)
- We want to reduce the % of available pores that are not sequencing (dark green). These pores are likely to become unavailable over time.

Legend

Sequencing
Pore currently
sequencing

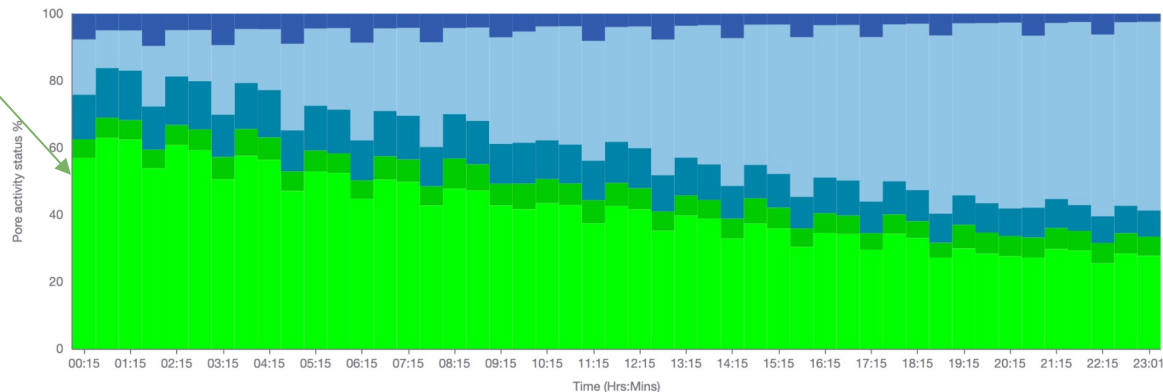
Pore available
Pore available for
sequencing

Unavailable
Pore currently
unavailable for
sequencing

Inactive
Pore no longer suitable
for further sequencing

Unclassified
Pore status unknown

Almost all
available pores
are sequencing
(light green) -
this is great!



Pore activity over time can also help indicate if the right amount of library was loaded

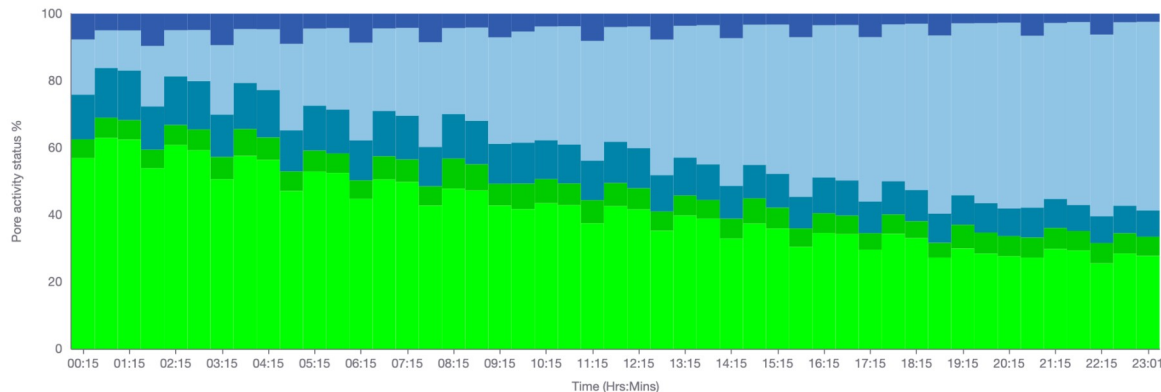
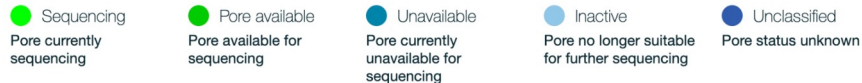
- Pore activity can provide information on how optimally the flow cell was loaded - **was the right amount of library loaded?**

^ PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

☒ Show grouped

Legend



Great loading!
Almost all available
pores are sequencing

Pore activity over time can also help indicate if the right amount of library was loaded

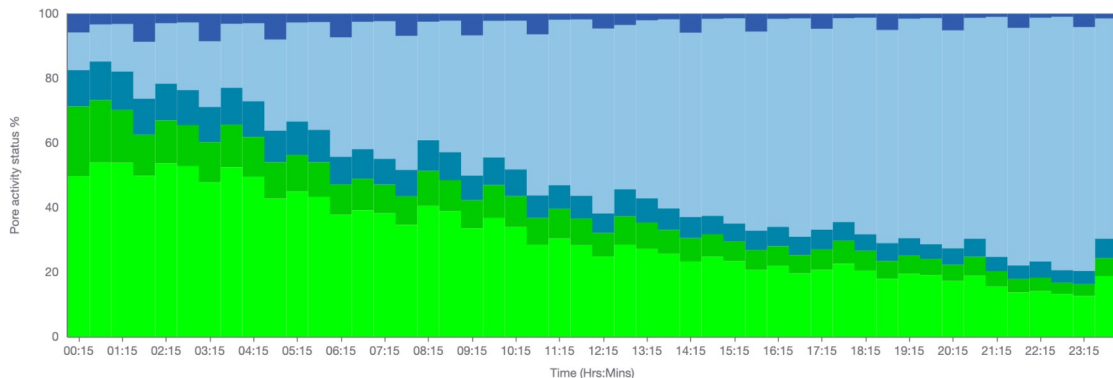
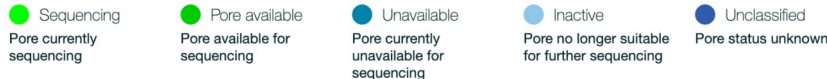
- Pore activity can provide information on how optimally the flow cell was loaded - **was the right amount of library loaded?**

^ PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

☒ Show grouped

Legend



Okay loading, but next time could load more fmol
There are still available pores (dark green)

Pore activity over time

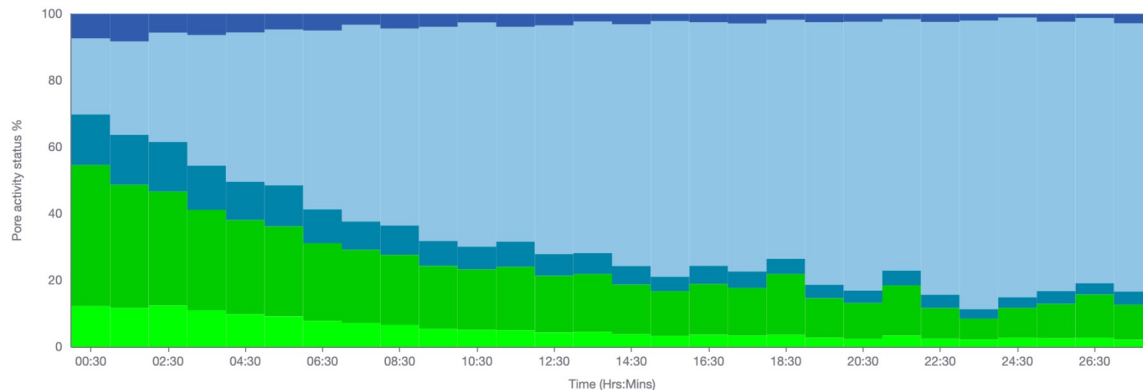
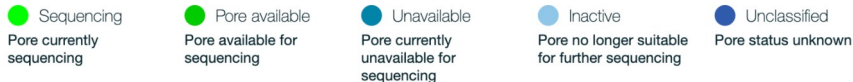
- Pore activity can provide information on how optimally the flow cell was loaded - **was the right amount of library loaded?**

^ PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

☒ Show grouped

Legend



Under loaded - there are too many pores available. Next time load more fmol of library

Run review exercise



Worksheet: Sequencing Run QC for ONT - Activity

- How much **output** (reads, bases) was generated?
 - *Hint: consider how many pores were available at the beginning of the run*
- How do the **read lengths** compare to expected / measured in the lab?
 - *Hint: compare to the Tapestation / QSep / diagnostic PCR report*
- How was the **barcode distribution** and how does this compare to expected based on sample pooling?
 - *Did your samples receive equal/expected number of reads/bases?*
- How was the pooling between the groups? Did each group receive the same amount of reads/bases?
- Do you think enough library was loaded?
 - *Hint: look at the pore activity chart*
- Metrics **over time** - any flags?
 - Output generated
 - Q-score
 - Pore activity and health

We recommend keeping a run log to track metrics over time.
This may help you identify abnormalities

Run review complete!

How do we prepare files for upload to CZ ID for analysis?

Next Slides: Prepare ONT fastqs for upload to CZ ID